

Planetary Orbit Visualisation — WebGL 2.0

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Project Direction

Scene Rendering System — a visually rich 3-D space scene with multiple celestial bodies, physically motivated lighting, and interactive camera control.

Idea Description

The application renders a gas giant and its icy moon in real time using a hand-written WebGL 2.0 / GLSL ES 3.00 pipeline (no game engine). The planet rotates on a tilted axis and is encircled by a semi-transparent ring system rendered as an alpha cutout. The moon orbits hierarchically and the two bodies exchange analytical shadows: the planet eclipses the moon on the far side and the moon casts a transit shadow dot on the planet when it crosses the sun. The user can orbit the scene freely with mouse/touch and scroll to zoom, and can toggle camera focus between the planet and the moon with a key press.

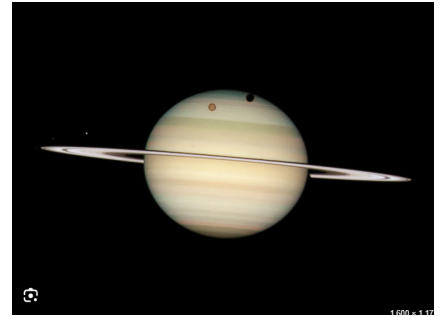
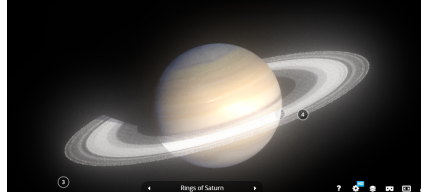
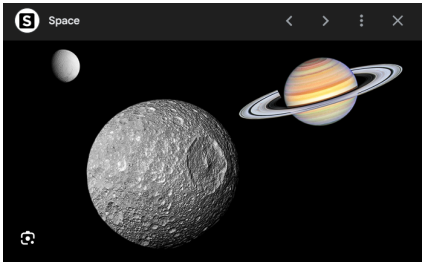
Baseline Techniques

Technique	Applies	Notes
Phong	Yes	
Material shininess & reflectance	Yes	
Diffuse texture maps	Yes	
Specular maps	Yes	
Per-fragment lighting	Yes	
Directional light	Yes	
Point lights with attenuation	No	Space scene has a single light source (the sun); no artificial point lights exist
Spotlights with smooth falloff	No	No cone-constrained lights applicable to this scene
Multiple light accumulation	No	Only one light source (directional sun); multi-light accumulation does not apply
GLSL uniforms & normal matrix	Yes	

Advanced Techniques

Technique	Effort	Implementation
Bloom	4/5	Bright-region extract pass → 6-pass separable Gaussian blur (ping-pong FBOs) → additive composite to screen
Environment Mapping	3/5	Star-field cubemap sampled via <code>reflect(-V,N)</code> per-fragment; contribution scaled by surface specular coefficient
Fog	2/5	Exponential deep-space fog; blends fragments toward a void colour with camera distance
Gamma Correction	2/5	Linear → gamma via <code>pow(max(col,0), vec3(1/2.2))</code> at end of shader
Combined	11/10	Requirement: $\geq 6/10$, at least one technique ≥ 3 ✓

Note: the selection of advanced techniques may be adjusted during implementation depending on feasibility and scope.



Left: Reference render 1. *Centre:* Reference render 2. *Right:* Reference render 3.

Inspiration Images

Implementation Plan

- **Step 1** — WebGL 2.0 context, VAO/VBO pipeline, per-fragment Phong shader with directional sun and 8K texture mapping.
- **Step 2** — GLB asset loading; scene-unit normalisation; hierarchical orbit transform with axial tilt and rings.
- **Step 3** — Cubemap skybox and environment mapping (per-fragment reflection vector sampling).
- **Step 4** — Bloom post-process (FBO pipeline); exponential fog; gamma correction.
- **Step 5** — Analytical inter-body shadows (ray-sphere test in fragment shader); view-toggle camera (planet $\leftrightarrow$ moon).
- **Step 6** — Polish: ring alpha cutout, anisotropic filtering, specular/gloss textures, performance tuning.

External Assets

Asset	Source	Licence
Planet & moon 3-D models	Sketchfab (BlendSwap mirror)	CC-BY
8K planet, ring & sun textures	solarsystemscope.com	CC-BY
Milky Way starmap cube-map	NASA / STScI — Gaia DR2 all-sky image	Public domain
<code>gl-matrix</code> (math library)	npmjs.com	MIT